

Improving the Robustness of LBP Face Recognition

ORL Face Dataset – Illumination and Rotation Robustness

Baseline

The baseline system uses uniform Local Binary Patterns, $\text{LBP}_{16,2}^{u2}$, with a 3×3 spatial histogram representation. Each probe image is matched to the gallery using the χ^2 histogram distance and nearest-neighbour classification. The fixed split uses images 1–5 of each subject as gallery images and images 6–10 as probes. The baseline performs well on normal images, but degrades under strong spatial illumination changes and larger rotations.

Improvement Method

Two complementary mechanisms address the baseline’s main weaknesses:

- **Illumination normalization and fusion.** Uneven lighting can make the same facial texture appear different to LBP. We use Gaussian smoothing ($\sigma = 8$) in the log-intensity domain to estimate broad shading, then subtract it to emphasize local facial texture. Rather than relying only on the corrected image, we combine the original and normalized LBP matching distances equally:

$$D_{\text{fused}} = 0.5D_{\text{raw}} + 0.5D_{\text{norm}}.$$

In simple terms, the matcher considers both *what the face looks like* and *what it looks like with uneven lighting reduced*.

- **Multi-angle rotation compensation.** When a face tilts, its eyes and other features shift across LBP’s fixed 3×3 regions. We therefore try rotating each probe back by several possible angles (-20° to 20° , every 5°) and keep the best match:

$$D_{\text{final}}(p, g) = \min_{\theta \in \{-20, -15, \dots, 20\}} D_{\text{fused}}(R_\theta(p), g).$$

This is like *straightening a tilted photo before comparing faces*; 0° is also tested for already aligned images.

Results

Baseline Accuracy		Final Robust Accuracy		
Condition	Accuracy	Condition	Final	Gain
Normal	0.975	Normal	0.975	0.000
Uniform brightness	0.975	Uniform brightness	0.975	0.000
Lighting ramp 0.2	0.960	Lighting ramp 0.2	0.975	+0.015
Lighting ramp 0.4	0.945	Lighting ramp 0.4	0.965	+0.020
Lighting ramp 0.6	0.895	Lighting ramp 0.6	0.975	+0.080
Lighting ramp 0.8	0.705	Lighting ramp 0.8	0.965	+0.260
Rotation 5°	0.970	Rotation 5°	0.965	-0.005
Rotation 10°	0.925	Rotation 10°	0.955	+0.030
Rotation 20°	0.630	Rotation 20°	0.880	+0.250

Summary. Normal-image accuracy is preserved at **97.5%**. Under the strongest illumination change, accuracy improves from **70.5% to 96.5%**; under a 20° rotation, it improves from **63.0% to 88.0%**. The only small regression is at 5° rotation (97.0% to 96.5%), while robustness improves substantially for the more difficult conditions. The main trade-off is higher computation because multiple rotations and two LBP branches are evaluated for each probe.